PAPER_567: Stellar Number Density Evolution n★(z) via Madau-Dickinson SFR

Unified Quantum Field Framework (UQFF)

Daniel T. Murphy • UQFF Research Consortium • 2026

Source: PAPER_567_Olbers_StellarDensityEvolution_nstar_z.md

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Author: Daniel T. Murphy — Star Magic / UQFF Framework **Session:** 153b **Gap-Fill For:** AldersOlbersBSFGMetricGapAnalysisCalculator (#160, PAPER_566) — Missing Extension 1 **Date:** 2026-03-29 **QS:** 5/5

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§1 Abstract

The classical Olbers' Paradox uses a constant stellar number density $n_\star$. In reality, the star-formation rate (SFR) evolves strongly with redshift: the universe was forming stars far more rapidly at $z \approx 2$ than today. This paper incorporates the **Madau-Dickinson SFR** into the UQFF 26-shell framework, replacing the constant $n_\star$ with a redshift-dependent $n_\star(z)$. The modified shell brightness is:

$$B_n = \frac{n_\star(z_n) \, L_\star \, \Delta r}{4\pi c (1+z_n)^4} \cdot R_{\mathrm{Ug1},n}$$

where $n_\star(z)$ grows steeply with z — paradoxically increasing the sky brightness at high redshift, but cut off by the DPM/VDS suppression before diverging.

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§2 Madau-Dickinson SFR

The cosmic star-formation rate density (Madau & Dickinson 2014):

$$\dot{\rho}_\star(z) = 0.015 \, \frac{(1+z)^{2.7}}{1 + \left[\frac{1+z}{2.9}\right]^{5.6}} \, M_\odot \, \text{yr}^{-1} \, \text{Mpc}^{-3}$$

Peak SFR at $z_\text{peak} \approx 1.9$:

$$\dot{\rho}_\star(z_\text{peak}) \approx 0.178 \, M_\odot \, \text{yr}^{-1} \, \text{Mpc}^{-3}$$

$$\dot{\rho}_\star(0) \approx 0.015 \, M_\odot \, \text{yr}^{-1} \, \text{Mpc}^{-3} \quad \text{(today)}$$

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§3 Stellar Number Density Evolution

Converting SFR to stellar number density via the main-sequence lifetime $\tau_\star$:

$$n_\star(z) = \frac{\dot{\rho}_\star(z) \cdot \tau_\star}{M_\star} \cdot (1+z)^3$$

where $(1+z)^3$ accounts for comoving volume compression:

$$\psi(z) = \frac{(1+z)^{2.7}}{1 + \left[\frac{1+z}{2.9}\right]^{5.6}}$$

$$n_\star(z) = n_{\star,0} \cdot \psi(z) / \psi(0) \cdot (1+z)^3$$

At $z = 2$: $n_\star(2) \approx n_{\star,0} \times 15.3 \times 3^3 = 413 \cdot n_{\star,0}$

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§4 UQFF DPM React Epoch Variation

Within the DPM 26-shell hierarchy, the vacuum reaction parameter DPM_react also varies with epoch (PAPER_516):

$$\text{DPM}_\text{react}(n) = \kappa_\text{DPM} \cdot \frac{\text{DPM}_n - \text{DPM}_s}{r_n} \cdot (1 + \delta_n)$$

where δ_n encodes the epoch-dependent aether state. At high z (shells 15–26), $\delta_n \sim \psi(z_n) - 1$ — the SFR excess above today.

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§5 Modified Olbers Integral

Shell brightness with evolving $n_\star(z)$:

$$B_n^\text{Madau} = \frac{n_\star(z_n) \, L_\star \, \Delta r}{4\pi c (1+z_n)^4} \cdot e^{-[\text{SSq}] \cdot n/26}$$

Compared to constant-density:

$$\frac{B_n^\text{Madau}}{B_n^\text{const}} = \frac{n_\star(z_n)}{n_{\star,0}} = \frac{\psi(z_n)}{\psi(0)} \cdot (1+z_n)^3$$

Shell	z_n	$\psi(z)/\psi(0)$	$(1+z)^3$	$n_\star(z)/n_{\star,0}$
1	0.128	1.56	1.43	2.2
5	0.641	3.71	5.00	18.6
13	1.666	9.84	34.0	334
20	2.564	9.71	107	1039
26	3.333	7.84	175	1372

Despite the large $n_\star(z)$ enhancement at high-z shells, the combined $(1+z)^4$ dimming and $e^{-[\text{SSq}]n/26}$ suppression keeps B_n convergent:

$$B_{26}^\text{Madau} \approx B_{26}^\text{const} \times \frac{1372}{(4.333)^4} \approx B_{26}^\text{const} \times 7.8$$

$$B_\text{sky}^\text{Madau} \approx 1.8 \times B_\text{sky}^\text{const}$$

The paradox remains resolved; SFR evolution provides a factor ~2 correction.

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§6 Faster Convergence at High-z

The SFR peak at $z \approx 2$ (shell 13) creates a local maximum in B_n, then the SFR declines at $z > 2$ — providing *faster convergence* beyond shell 13 than a pure power-law model would predict. This SFR turnover is a key observational feature of the resolved Olbers paradox.

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§7 Testable Predictions

1. **SFR peak feature:** Shell 13 ($z \approx 1.67$) contributes the most to B_sky; this corresponds to the cosmic star-formation peak — testable with JWST deep-field photon counts.
2. **EBL spectrum peak:** B_sky peaks in the optical/NIR (stellar emission), unlike a constant-$n_\star$ model.
3. **Factor ~2 correction:** UQFF predicts $B_\text{sky}^\text{Madau} \approx 1.8 \times B_\text{sky}^\text{const}$ — a measurable correction to PAPER_564 predictions.

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§8 Builds On / Addresses

Paper	Role
PAPER_564	DPM 26-shell Olbers (constant $n_\star$ — extended here)
PAPER_516	DPM shell energy including DPM_react
PAPER_566	Gap analysis — this is Missing Extension 1

PAPER_567 — Star Magic UQFF Framework — QS 5/5